

DEWAN NAFIUL ISLAM NOOR

Computer Science & Engineering | AI/ML Research | Software Development | Cybersecurity

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PROFESSIONAL PROFILE

I am a Computer Science and Engineering graduate from Bangladesh Army University of Engineering & Technology (BAUET) and currently pursuing an M.Sc. in Computer Science and Engineering at Rajshahi University of Engineering & Technology (RUET). I have a multidisciplinary background in Artificial Intelligence, Machine Learning, software development, cybersecurity, academic research, and technical project development. My experience includes AI/ML and computer vision research, programming and application development, penetration testing and vulnerability assessment, scientific writing, technical documentation, and collaborative project work. Alongside my technical and research experience, I have held several student leadership positions and developed strong skills in communication, teamwork, problem solving, project coordination, and academic presentation. I am interested in opportunities where I can apply my technical, analytical, research, and problem-solving abilities while continuing to grow professionally.

CORE EXPERTISE

Programming & Data: Python, C, C++, Java, JavaScript, PHP, NumPy, Pandas, Matplotlib, preprocessing, data analysis, experiment automation.

AI / ML / Computer Vision: Machine Learning, Deep Learning, HAR, TensorFlow, Keras, scikit-learn, OpenCV, CNN, LSTM/BiLSTM/GRU, ConvLSTM, Vision Transformer, VideoMAE, ResNet, VGG, ensemble learning.

Software / Web / Database: React/TypeScript exposure, Flutter, HTML/CSS, Bootstrap, MySQL, Supabase, XAMPP, Git/GitHub, Linux, desktop and database application development.

Cybersecurity: Vulnerability assessment, reconnaissance, network scanning, web/network security, mobile application security, Kali Linux, Nmap, Burp Suite, Metasploit, security reporting.

Research & Professional: Literature review, methodology design, cross-validation, benchmarking, visualization, scientific writing, LaTeX, technical presentation, documentation, mentoring, leadership.

EDUCATION

M.Sc. in Computer Science and Engineering - Rajshahi University of Engineering & Technology (RUET), Rajshahi | Apr 2026 - Expected 2028 | Currently pursuing

B.Sc. in Computer Science and Engineering - Bangladesh Army University of Engineering & Technology (BAUET), Natore | 2021 - 2025 | CGPA: 3.86/4.00

HSC - Government Azizul Haque College, Bogura | 2019 | GPA: 5.00/5.00

SSC - Bogura Zilla School, Bogura | 2017 | GPA: 5.00/5.00

RESEARCH & PROFESSIONAL EXPERIENCE

AI, Machine Learning & Computer Vision Research | 2024 - Present

- Conduct dataset preparation, preprocessing, experiment planning, model training, cross-validation, benchmarking, visualization, result interpretation, and manuscript preparation across video/wearable HAR, healthcare AI, computer vision, and low-light video understanding.
- Worked with HMDB51, UCF101, PAMAP2, IMDB-WIKI, UTKFace, and real medical-room video data; implemented/evaluated ConvLSTM, BiLSTM, GRU, CNN ensembles, Vision Transformer/VideoMAE-oriented pipelines, and hybrid temporal models.
- Co-designed methodology for the Springer Nature HAR study; contributed methodology and data collection plus research visualization/figure preparation to the PAMAP2 patient-monitoring study.

Penetration Testing Intern | Pentester Space, ICT Tower, Dhaka | May 2024 - Jun 2024

- Performed authorized vulnerability-assessment activities in controlled environments, including reconnaissance, scanning, vulnerability analysis, findings review, and security documentation.
- Used Linux-based security tools and gained practical exposure to web/network security assessment workflows, risk-oriented analysis, and structured technical reporting.

Team Matrix - Elite Hackers | Remote, Part-time | Feb 2021 - May 2021

- Participated in cybersecurity-focused practical learning involving account security, ethical hacking concepts, security tools, and structured problem solving in authorized contexts.

RESEARCH METHODS, DATA & SYSTEMS

Datasets: HMDB51, UCF101, PAMAP2, IMDB-WIKI, UTKFace, and real medical-room video data.

Research Workflow: Literature review, methodology design, preprocessing, experiment planning, model training, cross-validation, benchmarking, visualization, result interpretation, figure/table preparation, and manuscript writing.

Evaluation: Accuracy, precision, recall, F1-score, MAE, confusion-oriented analysis, train/validation/test workflows, comparative analysis, and reproducibility-focused experiment documentation.

Development Environment: Python, TensorFlow/Keras, scikit-learn, OpenCV, Git/GitHub, Linux, MySQL, PHP/JavaScript, React/TypeScript exposure, Flutter, Android Studio, Supabase, and XAMPP.

SELECTED TECHNICAL & RESEARCH PROJECTS

Ecommerce-Site-BD-Bazar | React, TypeScript, Supabase | Built a full-stack e-commerce, CMS, and mini-ERP application demonstrating modern web architecture, backend services, database workflows, and product-oriented development. [GitHub](#)

HAR-Activation-Optimizer-Study | Python, TensorFlow, OpenCV | Reproducibility-oriented ConvLSTM/BiLSTM HAR implementation using HMDB51/UCF101, multiple activation functions and optimizers, dataset preparation, training, and result comparison. [GitHub](#)

PAMAP2-CNN-BiLSTM-GRU-HAR | Python, TensorFlow/Keras, PAMAP2 | Wearable-HAR pipeline with preprocessing, windowing, scaling, 5-fold cross-validation, evaluation, streaming-related code, and TFLite export utilities. [GitHub](#)

Smart Face Recognition - Age & Gender Detection | Python, scikit-learn, NumPy, Pandas | Developed an image-based prediction workflow with preprocessing, feature handling, multiple ML models, comparative evaluation, and real-time/computer-vision progression. [GitHub](#)

Schedule Buddy - University Routine Management System | PHP, JavaScript, Bootstrap, MySQL | Led development of a responsive database-backed timetable system with administrative workflows and structured departmental scheduling. [GitHub](#)

University Academics Management System | Java, Swing/AWT, JDBC, MySQL | Built a desktop academic-management application with GUI, backend logic, database connectivity, and modules for students, faculty, attendance, leave, examinations, and fees. [GitHub](#)

Job Linker - Online Job Platform | Python, PHP, Flutter, Android Studio | Contributed to a cross-platform employment platform with backend/frontend components, job listings, category exploration, and skill-oriented job suggestions.

ADDITIONAL PROGRAMMING PORTFOLIO

OpenCV-Webcam-Viewer: Python/OpenCV webcam viewer with camera selection, mirror mode, preview, and screenshots.

Python Utilities: PDF Audiobook Reader, Jarvis Voice Assistant, and Desktop Auto-Reply projects demonstrating scripting, file processing, speech/utility workflows, debugging, and automation.

RESEARCH PUBLICATIONS & CONFERENCE WORK

Current research portfolio: 3 published papers, 2 IEEE conference papers successfully presented with proceedings/indexing pending, and 1 IEEE conference paper accepted for oral presentation.

1. **Effect of Activation Function and Model Optimizer on the Performance of Human Activity Recognition System Using Various Deep Learning Models** - *Discover Computing, Springer Nature, Vol. 29, Article 408, 2026* | Published | DOI
2. **Hybrid CNN-BiLSTM-GRU Architecture for PAMAP2-Based Human Activity Recognition in Patient Monitoring** - *Computer Science and Engineering Research, Vol. 3, No. 1, 2026* | Published | DOI
3. **Real-Time Face Recognition System for Age and Gender Prediction: A Robust CNN-Ensemble Approach with IMDB-WIKI and UTKFace Datasets** - *IEEE SPICSCON 2025, pp. 7-12* | Published | DOI
4. **PG-SpecTwin: A Multi-Task Deep Learning Framework for PCF-SPR Optical Sensor Design and Spectral Analysis** - *IEEE SPICSCON 2026* | Successfully Presented; Proceedings/Indexing Pending
5. **ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection** - *IEEE SPICSCON 2026* | Successfully Presented; Proceedings/Indexing Pending
6. **NightCare-VideoMAE: A Framework for Low-Light Patient Activity Recognition with Real Medical-Room Validation** - *IEEE IICAIET 2026* | Accepted for Oral Presentation; Camera-Ready/Publication Pending

RESEARCH & TECHNICAL HIGHLIGHTS

- Undergraduate thesis (Jul 2024 - Jul 2025): evaluated ConvLSTM, BiLSTM, Vision Transformer, ResNet34, VGG16, and VGG20 on selected HMDB51/UCF101 activity classes; compared ReLU, Sigmoid, and Tanh with SGD, Adam, RMSprop, and Adagrad.
- Model/evaluation experience includes train/validation/test workflows, 5-fold cross-validation, accuracy, precision, recall, F1-score, MAE, comparative analysis, model optimization, and reproducibility-focused experiment documentation.
- Research domains extend across patient monitoring, driver drowsiness detection, low-light activity recognition, age/gender analysis, intelligent video understanding, and AI-assisted optical-sensor analysis.
- Additional Python/GitHub work includes OpenCV Webcam Viewer, PDF Audiobook Reader, Jarvis Voice Assistant, and Desktop Auto-Reply automation utilities.

SELECTED RESEARCH CONTRIBUTIONS

- Springer Nature HAR study: co-designed the methodology for activation-function and optimizer analysis across recurrent deep-learning models using HMDB51 and UCF101.
- PAMAP2 patient-monitoring study: contributed to methodology and data collection and prepared research visualization/figures; the paper reports mean 5-fold accuracy of 99.50%.
- Published age/gender research used CNN/ensemble computer-vision methods with IMDB-WIKI and UTKFace; the paper reports approximately 97.8% gender-classification accuracy and 2.9-year age MAE.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Advanced Mobile Application Security Testing - Byte Capsule | Aug 2026
ISO/IEC 27001:2022 Lead Auditor - Mastermind Assurance | Dec 2025 - Dec 2028
Foundations of Log Analysis for Cyber Defense - Red Team Leaders | Nov 2025
Certified Cybersecurity Educator Professional (CCEP) - Red Team Leaders | Nov 2025
Introduction to Cybersecurity - Cisco Networking Academy | Nov 2025
Ethical Hacker (CEH); Kali Linux: Basic to Advanced - Team Matrix
EF SET English Certificate - EF Education First

LEADERSHIP, HONORS & ACADEMIC ENGAGEMENT

President, BAUET Computer Society (2025): Led technical activities, workshops, academic events, member coordination, and student engagement.
President, BAUET Welfare Club: Coordinated welfare/humanitarian initiatives and student/community engagement.
Campus Director, Hult Prize at BAUET (2024-2025): Managed team recruitment, communication, logistics, and entrepreneurship-event execution.
Additional Engagement: Core Organizer, Mind Storm 4.0 & 5.0; Publicity Secretary, BAUET Debating Society; member, BAUET Photography and Media Club.
Dean's Award - BAUET: Recognized for strong academic performance; also received leadership recognition for university organization presidency roles.

SECURITY TOOLS & WORKING ENVIRONMENT

Assessment / Learning Tools: Nmap, Burp Suite, Metasploit, John the Ripper, Hydra, Kali Linux, Linux command line, Windows, Git/GitHub, MySQL, and web-application development environments.
Governance / Monitoring: ISO/IEC 27001:2022 auditing fundamentals, log analysis, security controls, audit evidence, monitoring concepts, risk-oriented documentation, and cyber-defense fundamentals.

TEACHING & MENTORING CAPABILITY

- Can explain core CSE and AI concepts through practical examples, structured problem solving, laboratory tasks, and project-based learning; prepared to support programming, database, web-development, and machine-learning labs.
- Can guide student projects, research-oriented work, model evaluation, literature review, methodology description, experimental analysis, figures/tables, LaTeX formatting, technical presentations, and reports/manuscripts.

ACADEMIC, COMMUNICATION & PROFESSIONAL STRENGTHS

Academic / Mentoring: Prepared to support core CSE and AI/data learning, programming/database/web/ML labs, student projects, literature review, methodology writing, experimental analysis, figures/tables, LaTeX, and technical presentations.
Professional Strengths: Analytical problem solving, research collaboration, technical communication, documentation, scientific/technical writing, project coordination, team leadership, presentation, time management, and organized execution.
Relevant Coursework: Machine Learning, Digital Image Processing, Data Mining & Data Warehousing, Data Structures & Algorithms, Software Engineering, Database Systems, Computer Networks, Operating Systems, Network Security, Computer Graphics.

PROFILES & LANGUAGES

Profiles: LinkedIn | GitHub | Google Scholar | ResearchGate | ORCID
Languages: Bengali (Native) | English (Academic and professional working proficiency)

REFERENCES

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